Basic Algebraic Properties of Complex Numbers

Addition Multiplication

Closure

Adding or multiplying two complex numbers always gives another complex number. You never "leave" the set.

$z_1 + z_2 = \text{complex} \mid z_1 \cdot z_2 = \text{complex}$

Addition Multiplication

Commutative

Order doesn't matter — flipping the two numbers gives the same result.

$z_1 + z_2 = z_2 + z_1$

$z_1 \cdot z_2 = z_2 \cdot z_1$

Addition Multiplication

Associative

Grouping doesn't matter — you can bracket any two of the three numbers first.

$(z_1+z_2)+z_3 = z_1+(z_2+z_3)$

$(z_1 \cdot z_2) \cdot z_3 = z_1 \cdot (z_2 \cdot z_3)$

Additive identity

Zero does nothing

Adding $0 = 0 + 0i$ to any complex number leaves it unchanged.

$z + 0 = z$

Additive inverse

Negative cancels

Every complex number z has a negative $-z$. Adding them gives zero.

$$z + (-z) = 0$$

Multiplicative identity

One does nothing

Multiplying any complex number by $1 = 1 + 0i$ leaves it unchanged.

$$z \cdot 1 = z$$

Multiplicative inverse

Reciprocal exists (except for zero)

For every non-zero complex number z, there's a number w such that their product is 1. We call w the **multiplicative inverse**, written z^{-1}.

$$z \cdot w = w \cdot z = 1 \implies w = z^{-1}$$

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Commutative Property Under Addition

1 **Start — write z_1 and z_2 in standard form**

$z_1 = x_1 + iy_1 \quad z_2 = x_2 + iy_2$

where x_1, x_2, y_1, y_2 are real numbers

2 **Add them together**

$z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2)$

3 **Group real and imaginary parts**

$= (x_1 + x_2) + i(y_1 + y_2)$

4 **Swap order — real numbers can be added in any order**

$= (x_2 + x_1) + i(y_2 + y_1)$

because x_1, x_2, y_1, $y_2 \in \mathbb{R}$ and real addition is commutative

5 **Regroup back into complex form**

$= (x_2 + iy_2) + (x_1 + iy_1) = z_2 + z_1$

Conclusion

$z_1 + z_2 = z_2 + z_1$ ✓

Order doesn't matter when adding complex numbers.

Proof: Multiplicative Inverse

Finding z^{-1} for any non-zero complex number $z = x + iy$

1

Set up the inverse

Let $z^{-1} = u + iv$. By definition:

$z \cdot z^{-1} = 1$

→

2

Expand the product

Substitute and multiply out:

$(x+iy)(u+iv) = 1+0i$

$(xu-yv)+i(xv+uy) = 1+0i$

→

3

Match real & imaginary

Equate both sides:

$xu - yv = 1$

$xv + uy = 0$

→

4

Solve for u and v

Solve simultaneously:

$u = x/(x^2+y^2)$

$v = -y/(x^2+y^2)$

$z \neq 0 \Longrightarrow x^2+y^2 > 0$

→

5

Write the inverse

Substitute u and v back:

$z^{-1} = x/(x^2+y^2)$

$+ i\cdot(-y/(x^2+y^2))$

Not defined when $z = 0$

Key Takeaways

- Every **non-zero** complex number has an inverse: $z^{-1} = 1/z$
- Division of two complex numbers ($z_2 \neq 0$) is written as z_1 / z_2
- The inverse helps us **divide** complex numbers easily.

Complex numbers obey these formulas

i. $z^m \cdot z^n = z^{m+n}$

ii. $z^m / z^n = z^{m-n}$, $z \neq 0$

iii. $(z^m)^n = z^{mn}$

iv. $(z_1 z_2)^m = z_1^m \cdot z_2^m$

Thank You!